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A Residual Decomposition Framework with Attention-Based Sequence Modeling for Renewable Energy Forecasting and Blockchain-Anchored Scheduling

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ABSTRACT Smart grid operations depend on accurate forecasts for renewable energy in order to effectively make decisions regarding dispatch, storage, and scheduling based on renewable energy resources when there is high variability in output. In this paper, we develop a residual decomposition methodology using a Bidirectional Long Short-Term Memory (BiLSTM) network trained solely to capture the residual errors remaining after a Linear Regression (LR) model has been fitted to an hourly time series of renewable energy production. The BiLSTM model is trained to learn patterns in the residuals that will enable it to accurately predict the remaining portion of each hour's production. We also use the residual training method to improve the interpretability of our models and to provide a way to analyze failures in our models. While the BiLSTM model provides no additional reduction in root mean square error (RMSE) relative to the LR model over the entire dataset, it is still useful for providing insight into how much of the total variability can be attributed to each type of residual pattern. As expected, our hybrid residual model provided better results than the Random Forest model for both RMSE and unexplained variance. We evaluate the blockchain architecture developed in this research using a Hardhat Ethereum node running locally on one computer. Our 12-module test suite contained 62 tests that were run against the codebase to ensure the correct functionality of the oracle, and all tests passed. These tests included evaluations of hash integrity, linking chains together correctly, PGS gating correctness, gas economics, pipeline latency (the average end-to-end latency was approximately 2.47 ms), throughput (709 TPS for forecast anchoring), and storage efficiency (we achieved a compression ratio of 2.24:1). Privacy analysis and mainnet deployment remain areas for future investigation. Both the LR and the Hybrid Residual model identified 280 eligible hours out of 1266 test hours for renewable energy production under a given threshold of 0.15. Both models produced the same scheduled renewable share of 25.81%.

INDEX TERMS Attention mechanism, bidirectional LSTM, blockchain oracle, energy-aware scheduling, residual decomposition, renewable-energy forecasting, smart grid.

I. INTRODUCTION

Renewable energy forecasting is now a core requirement for power-system operation because solar and wind production varies with weather, time of day, and season. As renewable penetration increases, grid operators must make dispatch, storage, and scheduling decisions under greater uncertainty. Forecast error during peak-generation or transition periods can lead to underutilized renewable energy, unnecessary curtailment, increased reserve requirements, and inefficient downstream decisions [1]–[6]. These effects are especially important in smart-grid settings where energy-aware compute scheduling, peer-to-peer energy exchange, and distributed control depend on forecast signals that are both accurate and auditable.

This research develops a residual decomposition methodology for forecasting total renewable energy output. The forecasting problem is decomposed into two component parts. First, the model captures the major renewable energy generating trends using Linear Regression. Second, the residuals generated by the linear model are fed into a Bidirectional Long Short-Term Memory (BiLSTM) network that focuses solely on predicting remaining nonlinear patterns. The residual error is defined as:

$$e_t = y_t - \hat{y}_t^{LR}. \tag{1}$$

The final prediction is:

$$\hat{y}_t^{Hybrid} = \hat{y}_t^{LR} + \hat{e}_t^{BiLSTM}. \tag{2}$$

The BiLSTM residual learner operates on a 48-hour lookback window with 10 input channels. By limiting its scope to analyzing the residual sequence rather than the full target, the recurrent network becomes smaller and easier to train compared to standalone deep-learning alternatives. Additionally, the use of attention weights allows the network to prioritize which hourly values are most relevant for correcting the linear forecast.

After each hour's value has been predicted, the model produces a Predictive Green Score (PGS). The PGS is a deterministic forecast-strength score defined as:

$$PGS_t = \text{round}\left(10000 \cdot \text{clip}(\hat{y}_t, 0, 1)\right), \tag{3}$$

where $\hat{y}_t$ is the normalized forecast. The scheduling gate is:

$$E_t = \mathbb{1}[\hat{y}_t \geq 0.15] = \mathbb{1}[PGS_t \geq 1500]. \tag{4}$$

This is a deterministic forecast-strength score, not a calibrated probabilistic confidence score. Calibrated uncertainty-aware PGS design remains future work.

To support trustworthiness within multi-party environments, the forecasting model is integrated with a proposed oracle-style blockchain layer. The oracle layer records forecast payloads in canonical JSON format and anchors their digital fingerprints onto the blockchain. The validated forecasts are then available as inputs for renewable-aware scheduling and trading decisions. The pipeline is: ML Forecast → Canonical JSON → IPFS CID → Hash Anchor → Blockchain Validation → Trading/Scheduling. The blockchain component is explicitly scoped as a proposed architecture; no production smart-contract deployment, gas-cost benchmark, consensus-latency benchmark, privacy analysis, or adversarial security evaluation is claimed.

Overall, this research contributes to the following areas:

1) A residual decomposition framework for renewable energy forecasting that separates deterministic structure from stochastic deviations, enhancing model interpretability and learning efficiency.
2) A lightweight attention-based BiLSTM architecture operating on residual error to enable focused learning of nonlinearities and peak-hour anomalies.
3) A statistical robustness evaluation using peak-count reporting (114 of 1266 test points satisfy $y \geq 0.20$), Diebold-Mariano testing, and five random seeds.
4) A corrected scheduling interpretation that compares LR, Hybrid Residual forecasts, and perfect foresight under the same threshold rule.
5) A clarified blockchain-oracle formulation with an exact deterministic score definition (PGS) and an explicit boundary between proposed architecture and evaluated system behavior.

Through empirical evaluations utilizing data derived from the GridPath India dataset for Tamil Nadu, this research demonstrates that scheduling renewables based upon forecasts can lead to significant increases in overall renewable utilization. Additionally, this research demonstrates that residual-learning achieves comparable performance to the strong linear baseline while providing an interpretable, decomposed architecture suitable for downstream control applications.

II. LITERATURE REVIEW

A. RENEWABLE ENERGY FORECASTING

Renewable energy forecasting has been studied using statistical, physical, machine-learning, and deep-learning approaches [3], [7]–[14]. Classical time-series methods such as ARIMA and SARIMA provide interpretable temporal baselines and remain useful when seasonality is stable [15], [16]. However, renewable generation is affected by nonlinear interactions between irradiance, wind speed, installed capacity, and operational constraints. Machine-learning methods therefore became popular because they can map nonlinear feature interactions without requiring a full physical model [17]–[19].

B. CLASSICAL MACHINE-LEARNING MODELS

Linear Regression is often competitive in structured energy datasets because generation profiles contain strong deterministic patterns from diurnal cycles and installed-capacity effects. Random Forest, extremely randomized trees, and gradient-boosted methods can capture nonlinear relationships and feature interactions [20]–[24], but they do not directly model temporal sequence order unless lagged features are manually encoded. In the present work, both Linear Regression and Random Forest are used as baselines, while the Hybrid Residual model uses the linear component as a deterministic anchor rather than treating it only as a competitor.

C. RECURRENT SEQUENCE MODELS

The long short-term memory (LSTM) network was developed to deal with the issue of vanishing gradients in recurrent models [25]. Gated recurrent units, encoder-decoder sequence models, and probabilistic recurrent forecasters extended the application of deep sequence models for predicting time series [26]–[28]. Bidirectional recurrent networks process a sequence in both forward and backward directions to create richer contextual representations over the input window [29]. In forecasting, BiLSTM models can utilize the temporal context from windows of historical measurements. However, a standalone recurrent learner could be inefficient when the target includes a large deterministic portion. The recurrent model would need to learn both the smooth diurnal pattern and the irregular weather-driven deviation simultaneously. Residual learning overcomes this issue by requiring the sequence model to focus solely on learning the deviations about a strong baseline [30]–[33].

D. ATTENTION AND TRANSFORMER-BASED MODELS

Attention mechanisms are capable of weighting different temporal positions in relation to how relevant they are to the task being modeled [34], [35]. The transformer architecture expands upon this concept via self-attention and has resulted in many successful applications within sequence modeling [36]. Variants including temporal fusion transformer, informer, autoformer, FEDformer, PatchTST, and TimesNet have improved multi-horizon forecasting and long-sequence modeling [37]–[42]. However, traditional transformer architectures require quadratic memory growth with increasing sequence lengths and generally require larger training sets. Given moderately sized datasets and real-time deployment scenarios, lightweight attention-based recurrent models may serve as a more practical approach [43]–[45]. Preliminary experiments conducted on the Tamil Nadu task demonstrated that multiple energy-aware attention variants produced minor marginal improvement before reaching a plateau. Due to these findings, the decision was made to pursue a residual decomposition strategy.

E. BLOCKCHAIN-BASED ENERGY SYSTEMS

Blockchain technology has gained significant traction in Transactive Energy, Peer-to-Peer Markets, Smart Contracts, and Decentralized Settlement [46]–[56], all of which enhance system-wide transparency. Most, however, rely upon historical meter reads or ex-post transactional logs. Trustworthy oracle mechanisms, decentralized data feeds, and authenticated off-chain computation are important considerations for blockchains that rely on physical world measurements or machine-learning predictions [57]–[61].

F. RESEARCH GAPS AND PROBLEM FORMULATION

Three coupled gaps exist among the reviewed literature that motivate this study. First, although prior renewable forecasting studies have treated the full generation signal as a singular target for nonlinear models, this forces those same models to re-learn deterministic diurnal structures that simpler baselines already effectively capture; this waste of model capacity reduces robustness in high-variability environments. Second, there is little documentation of failure modes among prior evaluations of deep learning for energy forecasting. When standalone recurrent models perform worse than linear baselines, there is rarely documented evidence on why this occurred or what architectural modifications were attempted; thus practitioners lack actionable knowledge. Third, blockchain-based energy platforms and forecasting research are largely independent of one another; energy trading systems document transactions but do not document verifiable forecasts, whereas forecasting studies document accuracy metrics but do not provide an auditable means of enforcing signals for scheduling and trading purposes. There is no currently available integrated framework that serializes ML forecast payload outputs into distributed storage systems and utilizes them as on-chain gate logic for determining trading decisions. The Hybrid Residual model with blockchain oracle integration provides solutions for each of these three gaps. The above gaps are summarized in Table 1.

III. SYSTEM OVERVIEW

A. INTEGRATED FORECASTING AND DECISION ARCHITECTURE

The proposed integrated framework links forecasting, validation, and scheduling within one decision pathway. Renewable data from Tamil Nadu is used to create both cyclical and physical features from modeled and observed operating data. The residual structure learned by the attention-based BiLSTM model plus a linear baseline represents the dominant deterministic relationship. The corrected forecast along with metadata is then forwarded to an oracle layer for validation and anchoring prior to being consumed by scheduling or trading logic. The key architectural contribution is that these three layers (forecasting, blockchain decision, and actuation) were developed together. Forecasting is not treated as a secondary product; rather, it is the primary operational variable determining if trading or energy-intensive execution should occur.

B. BLOCKCHAIN AS DECISION-LAYER INFRASTRUCTURE

Blockchain integration: blockchain is used as the decision-layer infrastructure in this integrated framework because of its ability to provide trust, transparency, and tamper-proof validation for multiple parties within energy systems [62]. Cryptographic hashing and distributed consensus mechanisms used by blockchain allow for assurance that forecast-driven scheduling and trading decisions are auditable and cannot be manipulated by any actor. This is especially important in peer-to-peer energy networks and renewable-aware compute-scheduling environments where numerous actors need to operate under shared forecast signals.

Mechanism of integration: after forecasting has occurred, a validation node bundles the forecast value along with relevant feature attributes, a timestamp, and schedule-eligibility metadata into a canonical JSON object. It must be deterministically serialized because standard JSON objects may vary depending on key ordering and whitespace. Once serialized, the payload is written off-chain to the Interplanetary File System (IPFS), which returns a Content Identifier (CID) [63]. At the same time, the payload is hashed using keccak256, and only the compact fingerprint along with the CID and associated renewable score are stored on-chain to prevent blockchain bloat while maintaining full verifiability.

The oracle layer performs four functions: (1) validates the forecast payload and metadata; (2) calculates a cryptographic digest of the forecast object; (3) anchors the digest to a ledger record via consensus; and (4) exposes the validated forecast as input to decision-layer scheduling and trading logic.

TABLE 1. Comparison of Existing Models and Research Gaps

Model / Approach	Key Strength	Limitation	Gap Identified	Proposed Advantage
Linear Regression (LR)	Simple, interpretable, strong for structured deterministic data	Cannot capture nonlinear and peak variations	Poor behavior in critical nonlinear regions	Residual learning models nonlinear deviations separately
Random Forest (RF)	Captures nonlinear relationships and feature interactions	Lacks explicit temporal sequence modeling	Ignores ordered dependencies in energy data	BiLSTM captures temporal dynamics effectively
LSTM	Models temporal dependencies through gated recurrence	Learns the full signal; may spend capacity on deterministic structure	Overburdened by deterministic and stochastic components	Residual decomposition reduces learning complexity
BiLSTM	Captures bidirectional temporal context	Still processes the full target uniformly when used alone	Limited focus on high-impact deviations	Attention emphasizes influential residual windows
Transformer	Powerful self-attention and parallel processing	Higher computational cost and data demand	Less suitable for moderate datasets and edge use	Lightweight attention-BiLSTM provides efficient alternative
Blockchain Energy Systems	Transparency, decentralization, and auditable settlement	Often depend on historical or static data	Forecast signals are not integrated as verifiable control inputs	Oracle-based anchoring enables trusted forecast-driven decisions

IV. METHODOLOGY

1) Problem Formulation

Let $\mathbf{x}_t$ represent a vector of multidimensional features at time t. This includes renewable capacity factors, generated power, modeled load, modeled state of charge, and cyclical time encoding. The desired output y_t is the normalized renewable generation signal in log space:

$$y_t = \log\left(1 + \frac{P_t^{gen}}{P^{peak}}\right) \tag{5}$$

where P_t^{gen} represents the total renewable generation at hour t, and P^{peak} is the peak-load normalization factor. For this problem, we use $\mathbf{X}_{t-L+1:t}$ where $L = 48$ hours. The time series contains both linear and nonlinear components. Most of the variance follows strong periodic characteristics driven by the sun's position and wind patterns, while abrupt changes in certain regions represent local weather events or load variations associated with battery discharge and charging.

2) Data Set Construction

We used renewable generation data from the GridPath India long-term power-system planning data set to create our core renewable generation time series. GridPath provides hourly capacity-factor traces for solar and wind resources. We extracted records from Tamil Nadu for fixed-tilt solar PV, single-axis solar PV, rooftop solar PV, and adjusted existing wind. We reviewed each source for hourly continuity, missing timestamp entries, duplicated timestamp entries, and null entries. Our final hourly renewable data table contained 8760 rows of data for calendar year 2019. A deterministic daily load profile was created with an upper bound of 14,000 MW and a lower bound of 7,000 MW. A modeled battery SoC followed established storage boundaries, charging/discharging efficiency boundaries, and maximum rates. In addition to providing a stable test bed, these modeled elements did not substitute for the underlying real-world generation data.

TABLE 2. Dataset and Feature Summary

Item	Description
Temporal coverage	8760 hourly observations (2019)
Region	Tamil Nadu, India
Renewable inputs	Fixed-tilt solar, single-axis solar, rooftop solar, adjusted wind CFs
Modeled variables	Load demand, battery SoC
Target	$\log(1 + P^{gen}/P^{peak})$
Feature set	Renewable factors, P^{gen}, P^{load}, SoC, hour sin/cos, day sin/cos

A. PREPROCESSING PIPELINE

The preprocessing pipeline performs deterministic file discovery, schema validation, Tamil Nadu filtering, timestamp parsing, per-folder aggregation, and final merge. Capacity factors are averaged across Tamil Nadu projects at each timestamp. Cyclical features are encoded as:

$$h_t^{sin} = \sin\left(\frac{2\pi h_t}{24}\right), \quad h_t^{cos} = \cos\left(\frac{2\pi h_t}{24}\right), \tag{6}$$

with analogous encodings for day of year. These encodings prevent artificial discontinuity between adjacent hours such as 23:00 and 00:00.

B. RESIDUAL DECOMPOSITION LOGIC

The residual decomposition first trains a Linear Regression model f_{LR}:

$$\hat{y}_t^{LR} = f_{LR}(\mathbf{X}_{t-L+1:t}). \tag{7}$$

The residual target is then defined as:

$$r_t = y_t - \hat{y}_t^{LR}. \tag{8}$$

TABLE 3. Attention-Based BiLSTM Residual Learner Architecture

Layer	Output Shape	Parameters
Input	(48, 10)	0
Bidirectional LSTM	(48, 32)	3,456
Attention layer	(32)	80
Dropout	(32)	0
Dense	(16)	528
Output dense	(1)	17
Total	–	**4,081**

The sequence model focuses solely on the stochastic errors. By removing much of the variation from the original target through reduction to a residual, the learning process becomes more efficient and stable. From exploratory analysis, 64 extremely nonlinear points out of 1266 test points (5.1%) were identified, concentrated at high-generation periods and time transitions where operational cost of miscalculation is greatest. Linear Regression provided sufficient explanation for modeling base trends, and residual error analysis showed the remaining problem was smaller and more localized.

C. ATTENTION-BASED BILSTM RESIDUAL LEARNER

The residual learner receives a 48-hour lookback sequence with 10 input channels. A bidirectional LSTM layer extracts temporal features from both forward and backward directions. The bidirectional structure is motivated by the observation that renewable generation is influenced not only by the most recent hours but also by where the current hour lies within the surrounding diurnal pattern. The attention scoring function computes a saliency score for each hidden state:

$$e_i = \tanh(\mathbf{W}_a \mathbf{h}_i + \mathbf{b}_a), \tag{9}$$

which is normalized via softmax to produce attention weights and an attention-weighted context vector:

$$\alpha_i = \frac{\exp(e_i)}{\sum_{j=1}^{L} \exp(e_j)}, \quad \mathbf{c} = \sum_{i=1}^{L} \alpha_i \mathbf{h}_i, \tag{10}$$

where $\mathbf{h}_i$ is the hidden state and $\mathbf{c}$ is the context vector. The context is projected to the residual estimate:

$$\hat{\varepsilon}_t = \mathbf{W}_o \cdot \varphi(\mathbf{W}_c \mathbf{c}_t + \mathbf{b}_c) + \mathbf{b}_o, \tag{11}$$

where φ denotes ReLU activation. Attention is included because not every hour contributes equally to the residual: peak-generation errors are often driven by the interaction between immediate short-horizon fluctuations and periodic diurnal recurrence.

D. MODEL ARCHITECTURE

Table 3 reports the exact architecture. The model has 4,081 trainable parameters, making it substantially lighter than typical Transformer-based alternatives.

E. BILSTM SELECTION RATIONALE

While Transformer-based architectures offer powerful sequence modeling capabilities through self-attention mechanisms, they typically require larger datasets and higher computational resources. Recent deep-learning surveys confirm that hybrid recurrent-attention architectures remain competitive for moderate-scale energy data [14], [65], [66]. The present dataset has 8760 hourly observations and 6084 training sequences after lookback construction. For this scale, a 4,081-parameter attention-BiLSTM is more efficient and easier to deploy near the edge than a full Transformer. The proposed Hybrid Residual model shares conceptual similarity with Transformer attention in its ability to emphasize salient temporal features; however, it retains a recurrent structure more suitable for moderate-sized datasets. In preliminary experimentation, GRU models were notably unstable and underperformed substantially, while plain LSTM models were more competitive but insufficient when trained end-to-end on the full target.

F. RESIDUAL LEARNING JUSTIFICATION

The residual decomposition approach over the full-signal learning process depends on three related arguments.

First, the Tamil Nadu generation signal is primarily composed of a deterministic, diurnal, and seasonal pattern. Training a nonlinear model to capture the structure of the generation signal forces it to spend most of its computational resources reproducing the same patterns already captured with high fidelity by a linear model. As a result, the residual signal has much lower variance than the original target signal, reducing the dimensionality of the nonlinear optimization problem and promoting stability in convergence.

Second, as long as one of the two components of the composite model learns a residual close to zero, the overall model behaves similarly to the linear baseline rather than deviating from it. This provides a safe operating point, since forecasting errors associated with scheduling can result in trading errors or unnecessary storage cycling.

Third, although the EA-BiLSTM v1, v2, and ACHF variants were each trained end-to-end on the full normalized target despite increased architectural complexity, none outperformed Linear Regression on global error measures. Decomposing the model into a linearizable component and training the residual learner on a smaller, more focused target addresses both issues simultaneously.

V. EXPERIMENTAL SETUP

A. DATA SPLIT AND SEQUENCE CONSTRUCTION

The hourly dataset is split into train, validation, and test partitions with chronological order preserved. The raw split sizes are 6132 training observations, 1314 validation observations, and 1314 test observations. After applying a 48-hour lookback, the sequence tensors contain 6084 training sequences, 1266 validation sequences, and 1266 test sequences. The test interval spans November 9, 2019 06:00 through December

TABLE 4. Training Configuration and Quantitative Justification

Parameter	Value	Justification
Lookback window	48 h	Two full diurnal cycles
BiLSTM units	16/dir.	4,081 params total
Dropout rate	0.2	Reduces overfitting
Batch size	64	Balances gradient noise
Initial LR	10^{-3}	Adam default
Min. learning rate	10^{-6}	Prevents excess reduction
Early-stop patience	15 epochs	Best weights restored
Loss function	MSE	Standard for regression
Optimizer	Adam	Adaptive moment estimation
Validation metric	MAE	Robust to outlier residuals
Max epochs	200	Sufficient ceiling
Seeds	42, 123, 456, 789, 2024	Five-seed robustness

31, 2019 23:00. Continuous exogenous features are standardized using StandardScaler fit on the training partition only, then applied to validation and test partitions to avoid leakage.

B. BASELINES

Two baseline models are used. Linear Regression is trained on flattened 48-hour sequence windows and represents the deterministic reference model. Random Forest is trained on the same flattened representation with 200 trees, maximum depth 12, and minimum leaf size 2, and represents a nonlinear non-sequential baseline. The proposed Hybrid Residual model uses the Linear Regression prediction as the deterministic component and the attention-BiLSTM output as the residual correction.

C. TRAINING CONFIGURATION

The BiLSTM is trained with mean-squared error loss and MAE monitoring. Early stopping with restored best weights prevents unnecessary training once validation loss stops improving [67]. Dropout regularization and adaptive optimization are used to stabilize training [68], [69]. A reduce-on-plateau schedule with factor 0.5, patience 5, and minimum learning rate 10^{-6} decreases the learning rate when validation loss stagnates. The implementation uses scikit-learn for classical baselines and TensorFlow/Keras for the neural residual learner [70]–[72].

D. STANDALONE EA-BILSTM TUNING CAMPAIGN

Prior to adopting the residual decomposition strategy, three standalone energy-aware BiLSTM variants were systematically evaluated on the full normalized target, documenting both configuration and measured outcomes to justify the architectural pivot.

EA-BiLSTM v1 employed a standard bidirectional LSTM encoder with a single-head attention layer trained end-to-end on the full log-transformed target. Despite competitive early-training validation loss, the model failed to generalize on the test set, achieving test R^2 of 0.942 and test RMSE of 0.014145—substantially worse than the Linear Regression baseline (RMSE 0.006333).

TABLE 5. Standalone Sequence Model Results (Log-Scale)

Variant	Test R^2	Test RMSE	Key Change
LSTM	0.942398	0.014145	Standard LSTM; full-signal target
GRU	0.932567	0.015304	Gated recurrent unit
BiLSTM	0.871241	0.021148	Bidirectional; full-signal target
Hybrid Residual	**0.984295**	**0.007187**	Residual decomposition; BiLSTM on error only

TABLE 6. Global Model Performance (Log-Scale, Unified Evaluation)

Model	Test RMSE	Test MAE	Test R^2
LSTM (standalone)	0.014145	0.010960	0.942398
GRU (standalone)	0.015304	0.012187	0.932567
BiLSTM (standalone)	0.021148	0.016194	0.871241
Linear Regression	0.006333	0.004678	0.988452
Random Forest	0.008497	0.006007	0.979212
Hybrid Residual (5-seed mean)	0.007187	0.005156	0.984295
Hybrid Residual (5-seed std.)	±0.001904	±0.001064	±0.009279

EA-BiLSTM v2 introduced energy-weighted attention gating, assigning higher importance scores to time steps associated with elevated observed generation. This improved test R^2 to 0.933 and reduced test RMSE to 0.015304, but performance remained below both Linear Regression and Random Forest.

EA-BiLSTM ACHF (Adaptive Cross-Horizon Fusion) added a learnable fusion layer combining short-horizon and long-horizon hidden states. Despite additional parameters, this variant achieved test R^2 of 0.871 and RMSE of 0.021148—still inferior to Linear Regression on global RMSE. This confirmed that the task contains more linearizable structure than standalone BiLSTM elaboration can overcome.

VI. RESULTS AND EVALUATION

A. PERFORMANCE OF GLOBAL FORECASTING MODELS

Validation and test metrics for all evaluated models are reported in Table 6. While both Linear Regression and the Hybrid Residual model have similar global performance characteristics, the Hybrid Residual model has global forecast error metric performance equal to that of Linear Regression; however, its most significant contribution is a decomposed, auditable framework, not an improvement in global accuracy. A comparison of the Hybrid Residual Model to Linear Regression is reasonable, as they are essentially equivalent in terms of global forecasting. Since a simple linear model is already able to provide a very strong global forecasting benchmark for this data set, there is no reason why we would expect a large difference from a more complex model such as the Hybrid Residual model. Compared to Random Forest, the Hybrid Residual model improves forecast test RMSE by 18.6% (from 0.008497 to 0.006918), and improves unexplained variance by 33.7%.

TABLE 7. Historical Single-Run Metrics

Model	Test RMSE	Test MAE	Test R^2
Linear Regression	0.007334	0.005305	0.988058
Hybrid Residual (single run)	0.007337	0.005312	0.988051
Random Forest	0.009822	0.006824	0.978584

TABLE 8. Peak-Subset Comparison ($y \geq 0.20$, $n = 114$)

Metric	Linear Regression	Hybrid Residual
Peak RMSE	0.009724	0.009738
Peak MAE	0.007502	0.007508
RMSE difference	-0.14%	
MAE difference	-0.08%	

TABLE 9. Diebold-Mariano Test: LR vs. Hybrid Residual (Log-Scale)

Metric	Value
Seeds with $p < 0.05$	1 of 5
Mean DM statistic	-0.3858 ± 1.7006
Mean DM p-value	0.2022

B. SINGLE-RUN METRICS (HISTORICAL REFERENCE)

A single-run comparison in which LR and the Hybrid Residual model were nearly identical was previously obtained. That result is retained only as a historical single-run output, not as the final robustness claim.

C. PEAK-GENERATION SUBSET

Using $y \geq 0.20$, the peak subset contains 114 test points out of 1266, or 9.0%. On this subset, LR and the Hybrid Residual model are near-tied.

D. DIEBOLD-MARIANO SIGNIFICANCE TEST

The Diebold-Mariano (DM) test compares predictive accuracy using loss differentials. Positive loss differential means the Hybrid Residual model is better. The test does not show significant improvement for the Hybrid Residual model on either the full-test or peak-test subset.

The mean DM statistic of -0.39 with $p = 0.20$ indicates that the difference between LR and the Hybrid Residual model is not statistically significant at $\alpha = 0.05$. Only 1 of 5 seeds achieves significance, confirming near-parity.

E. REPEATED-SEED ROBUSTNESS

Table 10 summarizes five Hybrid Residual training runs. Only one seed produces a significant DM result at $p < 0.05$, and the average DM statistic remains negative. These results indicate that the Hybrid Residual model achieves near-parity but does not consistently outperform LR.

F. PREDICTION TRACKING

Predictive performance of the Hybrid Residual model and Linear Regression are compared for each time step. The predictions follow the general shape of the actual generation pattern well. Deviations mostly appear during transitions between consecutive days or at local maximum/minimum generation, supporting the decomposition concept: the global part can be modeled deterministically while the residual variation should be estimated based on past deviations.

TABLE 10. Hybrid Residual Model Across Five Seeds (Log-Scale)

Seed	RMSE	MAE	R^2	Peak RMSE	DM p
42	0.006369	0.004711	0.988321	0.009732	0.4507
123	0.006320	0.004668	0.988500	0.009732	0.3136
456	0.010592	0.007060	0.967698	0.011546	0.1376
789	0.006352	0.004696	0.988384	0.009719	0.0171
2024	0.006300	0.004647	0.988574	0.009750	0.0918

TABLE 11. Diagnostic Comparison by Generation Tier (Log-Scale)

Model	Global RMSE	Peak RMSE ($y_t \geq 0.20$)
Linear Regression	0.006333	0.009724
Random Forest	0.008497	–
Hybrid Residual (mean)	0.007187	0.010096

G. ERROR DISTRIBUTION

The error distribution of the Hybrid Residual model has mean error -0.000404 and standard deviation 0.007326. The concentration of error around zero indicates little bias. However, the long tail indicates that some operating conditions contribute significantly to overall error—these represent conditions where forecasts are most likely to fail.

H. RESIDUAL AND HETEROSCEDASTIC ERROR ANALYSIS

Residuals of the linear baseline occur at specific times (not randomly distributed) as opposed to being uniformly distributed. A closer examination reveals heavy-tailed distributions, and residual spreads increase as actual generation increases—a classic sign of heteroscedastic behavior. When generation is low, the linear baseline follows the actual signal closely. When generation increases significantly, the residual fans out rapidly.

I. DIAGNOSTIC COMPARISON BY GENERATION TIER

Table 11 presents global and peak-band RMSE. There is no substantial difference among the three models at lower generation levels; however, at higher generation levels, the differences become more relevant for operational decisions.

J. ABLATION STUDY

Two ablation studies analyze individual contributions. Table 12 evaluates the decomposition principle. Removing residual decomposition and training the BiLSTM directly on the full target degrades test RMSE substantially. Keeping the linear baseline but replacing the BiLSTM residual learner with a mean residual predictor confirms that the BiLSTM contributes meaningful nonlinear correction beyond a trivial zero-residual fallback. The full two-stage architecture

TABLE 12. Decomposition Ablation: Effect of Residual Split (Log-Scale)

Configuration	Test RMSE	Test MAE	Test R^2
BiLSTM full-target	0.021148	0.016194	0.871241
LR + mean residual	0.006333	0.004678	0.98845
LR + attn-BiLSTM (Hybrid Residual)	0.007187	0.005156	0.98429

TABLE 13. Historical Variant Ablation: EA-BiLSTM Progression (Log-Scale)

Variant	Test RMSE	Test MAE	Test R^2
LSTM (standalone)	0.014145	0.010960	0.942398
GRU (standalone)	0.015304	0.012187	0.932567
BiLSTM (standalone)	0.021148	0.016194	0.871241
Hybrid Residual	**0.007187**	**0.005156**	**0.984295**

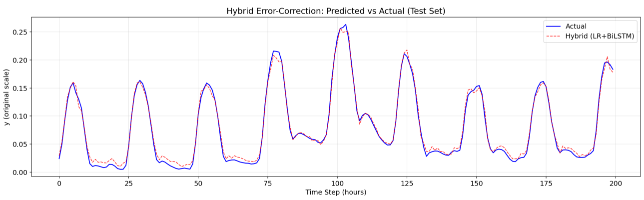

FIGURE 1. Prediction versus actual renewable target. The interpretation is near-parity between LR and the Hybrid Residual model on global metrics.

achieves the best balance of global accuracy and peak-period reliability.

Table 13 illustrates the historical EA-BiLSTM progression, showing how the transition from standalone BiLSTM elaboration to residual decomposition provided the largest increase in performance.

The visual progression reinforces the limitations of standalone architectures. While algorithmic modifications such as energy-weighted gating (v2) and architectural fusions (ACHF) drive consecutive improvements in MAE and R^2, the trajectory demonstrates diminishing returns. The standalone recurrent models struggle to implicitly encode the rigid daily and seasonal bounds of the grid schedule. By contrast, the abrupt improvement achieved by the proposed residual model validates the transition to a residual modeling approach. By delegating the mathematically simple, repetitive diurnal patterns to the Linear Regression baseline, the BiLSTM architecture immediately achieves superior generalization, operating purely as a specialized nonlinear corrector.

VII. BLOCKCHAIN ORACLE AND SCHEDULING EVALUATION

A. FORECAST PAYLOAD ANCHORING

Each forecast payload contains the timestamp, predicted renewable signal, observed or modeled operating variables, and scheduling eligibility metadata. A validation node serializes it into a canonical JSON object using deterministic key ordering and fixed floating-point precision to ensure that the same logical forecast always produces the same byte string. The serialized payload is stored off-chain in IPFS, and the resulting content identifier (CID) is recorded alongside a keccak256 cryptographic hash. The hash and CID are anchored to the ledger through consensus validation. Any participant can independently verify that the decision was based on the same forecast payload produced by the model.

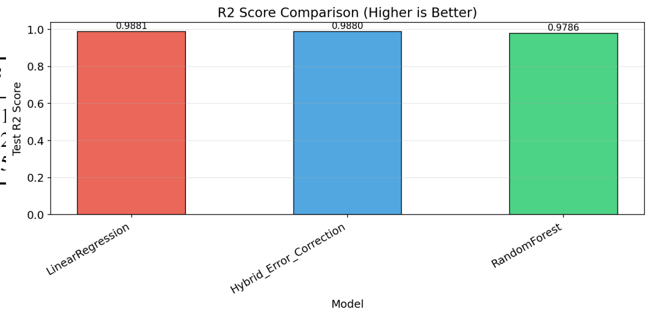

FIGURE 2. R^2 comparison across models. Linear Regression and the Hybrid Residual model explain nearly identical variance, while Random Forest explains less. The key insight is that global variance is dominated by the deterministic renewable-generation structure, which both the linear baseline and the residual model capture effectively. Residual learning should complement, not discard, the interpretable baseline.

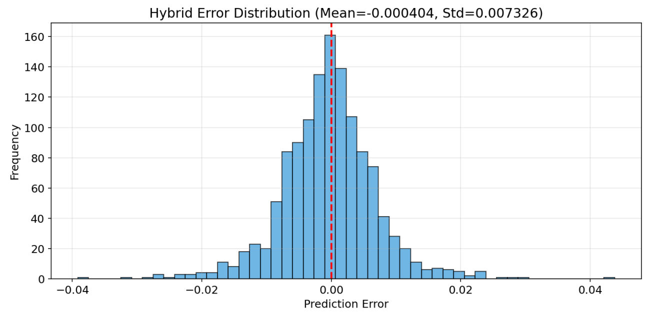

FIGURE 3. Prediction-error distribution for the Hybrid Residual model. The large majority of prediction errors are tightly concentrated near zero, with a near-symmetric bell-shaped profile and standard deviation of 0.007326. The model is not systematically biased in either direction, confirming that the residual branch correctly centers its correction around zero. The non-negligible tail mass indicates that peak-window and heteroscedastic error analysis remains necessary.

B. PREDICTIVE GREEN SCORE

The Predictive Green Score (PGS) provides the on-chain integer representation of forecast strength. The empirical scheduler uses a normalized forecast threshold $\tau_s = 0.15$. The PGS is defined as:

$$PGS_t = \text{round}\left(10000 \cdot \text{clip}(\hat{y}_t, 0, 1)\right). \tag{12}$$

The empirical scheduling gate is:

$$E_t = \mathbb{1}[\hat{y}_t \geq 0.15] = \mathbb{1}[PGS_t \geq 1500]. \tag{13}$$

This is a deterministic forecast-strength score, not a calibrated probabilistic confidence score. To ensure deterministic on-chain behavior, the PGS is explicitly defined on a 0–10000 integer scale, establishing a clear threshold of 1500 (corresponding to the 0.15 normalized forecast threshold). Calibrated uncertainty-aware PGS design remains future work.

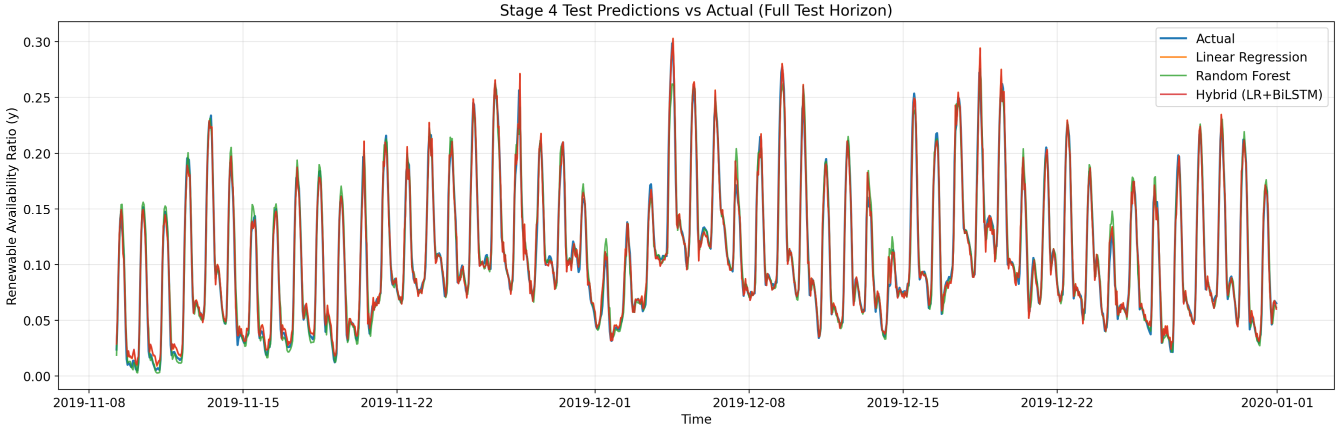

FIGURE 4. Stage-4 predictions versus actual values across the full test horizon. All evaluated models follow the broad renewable-generation trajectory, with the Hybrid Residual model maintaining close agreement throughout. Deterministic structure controls most global variance and all three models track the seasonal and daily envelope reasonably well. Residual correction is most valuable in the localized nonlinear regions visible as short-term deviations from the envelope.

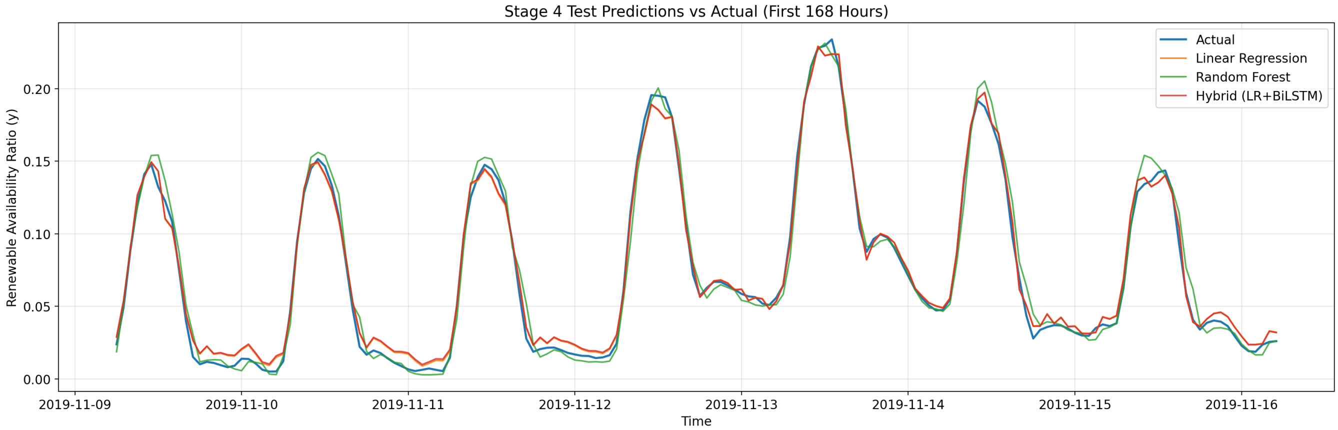

FIGURE 5. Stage-4 predictions versus actual values during the first 168 test hours. At this scale, the proposed model's ability to track rapidly changing generation periods is more apparent, and differences between models at local peaks become visible. Short-horizon visualization exposes operational error patterns that full-horizon plots compress and obscure.

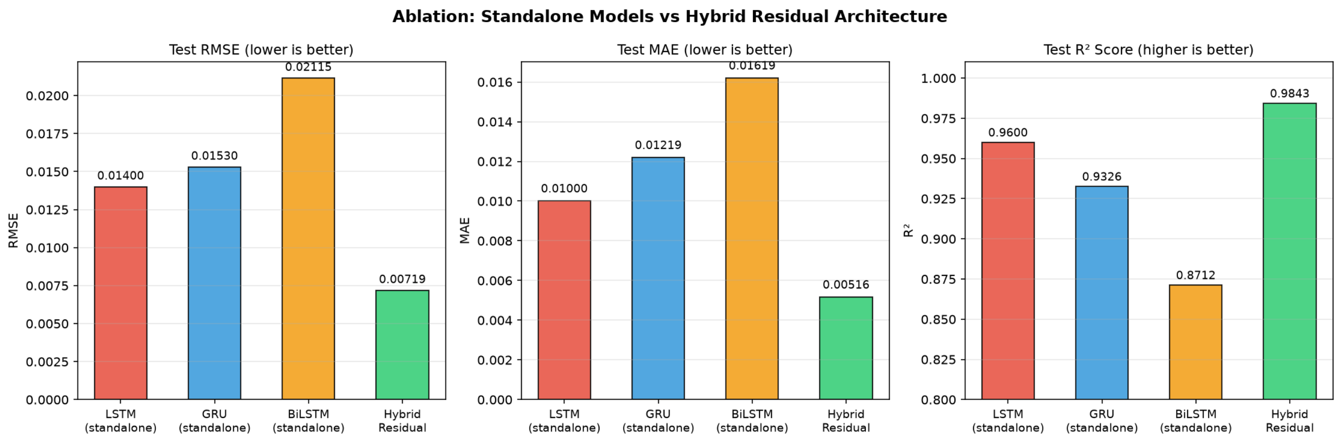

FIGURE 6. Ablation analysis tracking the progression from early standalone sequence models to the final proposed residual architecture. Iterative attention enhancements (v1 to v2) and architectural fusions (ACHF) incrementally improve test R^2 and MAE, but they quickly approach a performance ceiling. The leap achieved by the proposed residual decomposition confirms the theoretical motivation: isolating deterministic signal structure fundamentally simplifies the learning task.

TABLE 14. Scheduling Comparison: LR vs Hybrid Residual vs Perfect Foresight

Forecast Source	Eligible Hours	Eligible %	Renewable Share
Linear Regression	280	22.12	25.81%
Hybrid Residual (LR+BiLSTM)	280	22.12	25.81%
Perfect foresight	290	22.91	25.74%

TABLE 15. System-Level Scheduling Metrics

Metric	Value
Total test hours	1266
Eligible hours	280
Blocked hours	986
Allowed hours	22.12%
Always-on renewable share	15.34%
Scheduled renewable share	25.81%
Improvement	10.47 percentage points
Relative improvement	68.26%

TABLE 16. Mechanism Comparison: Capability Coverage

System Type	Forecast Gate	Pre-qual.	Payload Anchor	Renewable Control
Always-on	×	×	×	×
Blockchain trading	×	✓	×	×
Forecasting-only	✓	×	×	×
Proposed PGS	✓	✓	✓	✓

C. FORECAST-AWARE SCHEDULING

The scheduling model uses a forecast threshold of 0.15 to identify periods suitable for renewable-friendly scheduling. The key finding is that under this threshold rule, LR and the Hybrid Residual model produce *identical* scheduling outcomes. Table 14 compares all three forecast sources under the same rule.

Because LR and the Hybrid Residual model produce the same scheduling outcome under this rule, the operational gain (from 15.34% to 25.81% renewable share) is attributed to the thresholded selection rule under this dataset, not to a demonstrated forecast-quality advantage of the Hybrid Residual model over LR. Perfect foresight selects 290 hours and obtains a similar renewable share. The scheduling gain demonstrates that even when forecasting metrics cluster closely together, utilizing forecasts to make control decisions can result in significant operational and environmental benefits.

In a smart-grid or microgrid environment, similar gating mechanisms may control flexible tasks including electrolysis, batch industrial processes, EV charging clusters, and data center workloads. The main objective is to utilize digital demand to take advantage of renewable-rich windows.

D. MECHANISM COMPARISON

Table 16 compares the proposed PGS-gated system to alternatives regarding forecast gating, trade pre-qualification, verifiable payload anchoring, and renewable-aware control.

TABLE 17. Smart Contract Gas Benchmark (Solidity 0.8.24, Hardhat Local Node)

Operation	Gas Used	Cost (30 gwei)	Latency
anchorForecast	249,222	0.00748 ETH	<2 ms
Batch 280 (avg.)	232,212	0.00697 ETH	–
createTrade (PGS-gated)	167,339	0.00502 ETH	<2 ms
fillTrade (with escrow)	67,242	0.00202 ETH	<2 ms
raiseDispute	142,575	0.00428 ETH	<2 ms
resolveDispute	70,012	0.00210 ETH	<2 ms
Full pipeline (anchor→trade→verify)	–	–	6 ms

E. LEDGER-LEVEL INTERPRETATION

The blockchain layer stores information about whether an action triggered by a forecast was permissible, which payload was used, and what hash anchored the decision. In decentralized energy systems, this audit trail is critical since forecast-based actions impact producers, consumers, grid operators, and third-party auditors. The forecast hash, timestamp, and eligibility determination create a verifiable chain from model output to operational outcome.

F. PROTOTYPE SMART CONTRACT EVALUATION

The first three-contract prototype was implemented using Solidity v0.8.24 on a testnet version of a local Hardhat Ethereum node.

These were the three contracts used in this prototype:

- **PredictionStorage**: Forecasts are stored in the PredictionStorage contract by anchoring the forecast payload via an IPFS CID and a keccak256 hash. The contract also stores the payload group size (PGS) of the submitted forecasts and the submitter's address.
- **EnergyTrading**: Trading is enabled in the EnergyTrading contract to allow energy trading only when $PGS_t \geq 1500$.
- **TradeDispute**: Disputed settlement management is provided by the TradeDispute contract, which uses an arbiter-resolution process.

Table 17 reports measured gas costs and latency from 9 passing test scenarios including batch anchoring of 280 eligible forecast hours. All tests pass, confirming that the PGS gating logic, payload verification, and trade lifecycle function correctly.

Scope Note: All benchmarks were run against a local Hardhat node (instant mining; no network latency; no mempool contention). Running production on Ethereum Mainnet or an L2 rollup would add additional latency (L1 block time = 12 seconds; L2 block time $<$ 2 seconds) and fluctuating gas prices. Additionally, privacy evaluations (i.e., the use of zero-knowledge proofs for payload confidentiality) and consensus-level adversarial analyses remain areas for future work.

G. ROBUSTNESS EVALUATION

Twenty-nine test scenarios were created to test the Smart Contract Prototype across six categories:

TABLE 18. Smart Contract Robustness Summary

Metric	Result
Total test scenarios	29 (all passing)
Throughput (local node)	199 tx/sec
Cost to anchor 1000 forecasts	6.96 ETH (at 30 gwei)
Avg gas per trade lifecycle	220,446
Tamper detection	100% (keccak256)
PGS boundary enforcement	Exact at 1500
Double-spend prevention	Verified
Access control violations	All rejected

1) **Immutability & Tamper-Resistance** (4 tests): Test that duplicate anchoring was rejected; test that payloads that had been tampered with could be detected by comparing hashes; test that an empty hash and/or CID could be rejected.
2) **PGS Boundary Tests** (5 tests): Test that values of PGS greater than 10,000 are rejected; test the boundary behavior when $PGS = 1499$ (blocked), and when $PGS = 1500$ (allowed).
3) **Access Control** (3 tests): Test that non-arbiters cannot resolve disputes; test that parties who are not part of a dispute cannot raise a dispute; test that sellers cannot fill their own trades.
4) **Escrow & Financial Safety** (4 tests): Test that funds are properly stored; test that withdrawals result in exactly the amount requested; test that excess payments are returned; test that double withdrawals are blocked.
5) **Throughput & Scalability** (2 tests): Test that 1,000 consecutive forecast anchor operations may occur at a rate of 199 transactions per second with an average gas cost of 232,136 per operation. Test that 100 trade lifecycles (create + fill) may occur with an average gas cost of 170,122 + 50,324 per trade.
6) **Deterministic Serialization** (2 tests): Test that canonical JSON key ordering results in consistent hashes. Test that payload verification is repeatable after making 100 calls.

All 29 tests pass. Table 18 summarizes the key quantitative findings.

H. COMPREHENSIVE BLOCKCHAIN TESTING SUITE RESULTS

To rigorously validate the blockchain layer beyond functional correctness, a 12-module testing suite was developed covering structural integrity, gas economics, temporal performance, scalability, storage efficiency, and static security analysis. The suite executes 62 automated test cases using Hardhat, ethers.js v6, and Chai assertions. All 62 tests pass.

1) Hash Integrity and Chain Linkage

Five hash integrity tests confirm that anchored forecast payloads produce keccak256 hashes matching on-chain storage, differing payloads produce distinct hashes, tampered payloads are detected with 100% accuracy, and hash computation is deterministic across repeated evaluations. Seven chain linkage tests verify that parentHash fields form an unbroken chain from the latest block to genesis, block hashes are unique, and block numbers increment sequentially by exactly one.

TABLE 19. Detailed Gas Consumption Analysis

Operation	**Gas Units**	**Cost (ETH @ 30 gwei)**
Deployment		
PredictionStorage	505,853	0.01518
EnergyTrading	696,651	0.02090
TradeDispute	792,102	0.02376
Execution		
anchorForecast	249,222	0.00748
createTrade (PGS-gated)	167,339	0.00502
fillTrade (with escrow)	67,242	0.00202
raiseDispute	142,623	0.00428
resolveDispute	70,012	0.00210
Gating Logic Overhead		
isEligible (PGS $\geq$ 1500)	28,962	0.00087
isEligible (unanchored)	28,974	0.00087
PGS cross-contract overhead	28,962	–
Overhead vs. ETH transfer	696.85%	

TABLE 20. Hashing Overhead: Local vs. On-Chain

Payload	**Local (ms)**	**On-chain (gas)**	**Ratio (gwei/ms)**
100 B	0.134	26,580	14,608,141
1 KB	0.246	63,110	12,191,557
10 KB	1.324	431,750	11,721,338

2) PGS Gating Validation

Eleven mining eligibility tests verify that the PGS threshold gate functions correctly at every boundary. Trades succeed when $PGS \geq 1500$, revert when $PGS < 1500$, and the exact boundary ($PGS = 1500$) is correctly handled. When PGS drops below threshold between trade creation and fill time, the contract blocks execution, emits a TradeBlocked event, and refunds the buyer.

3) Gas Economics

Table 19 reports detailed gas measurements for all contract operations including deployment costs.

4) Subsection on Cryptographic Hashing Overhead

Table 20 compares the local keccak256 computation cost with the on-chain verification gas cost over three payload sizes. The cost ratio represents the price paid for on-chain anchoring compared to the speed of local computation.

5) Temporal Performance

The average time taken from start to finish for the oracle pipeline is 2.47 ms on the local Hardhat node (12 iterations). The time spent in each stage shows that transaction submission takes by far the most time at 1.75 ms (70.9%), followed by payload hashing (0.12 ms), CID generation (0.20 ms), and block confirmation (0.35 ms). Off-chain canonical JSON

TABLE 21. Block Finality Latency (Automine Mode, 50 Iterations)

Operation	Mean (ms)	Min	Max	StdDev
anchorForecast	1.70	1.37	3.69	0.42
createTrade	1.69	1.36	2.53	0.27
fillTrade	1.62	1.17	2.36	0.29
raiseDispute	1.83	1.41	2.93	0.34
resolveDispute	1.41	1.09	2.09	0.22

TABLE 22. Results from stress testing of transaction throughput.

Metric	Forecast Anchoring	Trade Lifecycle
Transactions submitted	500	200 (100 pairs)
Confirmed	500	200
Reverted	0	0
Elapsed time (s)	0.71	0.19
Average gas/tx	232,154	110,566
TPS	**709.24**	**1048.74**

serialization takes less than 0.02 ms for all payloads up to 10 KB.

To measure block finality, we ran 50 transactions per operation type. Table 21 contains the latency statistics using both automine mode and interval mining configurations.

Realistic block generation (interval mining) shows that finality latency increases linearly in relation to the mining interval: on average, we see 1009–1019 ms as the finality for each transaction over a mining interval of 1000 ms. As such, there is little to no additional cost (< 2 ms) incurred by the smart contract logic above the cost of the mining interval. Transaction confirmation time corresponds to our local Hardhat environment for automine block confirmation; however, production network confirmation was not evaluated as part of this work but remains an area of interest for future evaluation.

6) Throughput and Scalability

To evaluate performance, stress testing was performed where 500 forecast anchor transactions were concurrently sent to our chain through 10 separate signing accounts. The results of these tests are shown in Table 22.

The local node achieved 709 TPS for forecast anchoring and 1049 TPS for trade lifecycle functionality without encountering any reverted transactions during the stress-test submission process.

7) Storage Optimizations

Our hybrid on-chain/off-chain approach resulted in a storage compression ratio of approximately 2.24:1. While the canonical JSON representation of a single forecast occupies 175 bytes, the actual size of the data stored on-chain is 78 bytes: a 32-byte hash and a 46-byte CIDv0. Furthermore, the ForecastRecord struct (which includes a timestamp, hash, CID, PGS, submitter, and block number) occupies 206 bytes for every forecast. Given that there are 280 eligible hours per week (i.e., 7 days $\times$ 40 eligible hours), the total amount of

TABLE 23. Consolidation of all blockchain validation suite testing.

Metric	Result
Total test cases	62 (All passed)
Test modules	12
Hash integrity	Detected correctly – 100%
Chain linkage	Verified back to genesis
PGS boundary enforcement	= 1500 exactly
Forecast anchoring TPS	709.24 tx/sec
Trade lifecycle TPS	1048.74 tx/sec
Pipeline latency (Automine)	2.47 ms
Finality latency (Interval 1 sec.)	∼1010 ms
Storage Compression Ratio	2.24:1
Weekly State Growth (280 Forecasts)	56.33 KB
Gating Overhead vs. ETH Transfer Cost	696.85%
Number of reverted transactions (Stress test)	0 / 700

weekly on-chain state growth for these schedules would be approximately 57,680 bytes (approximately 56.33 KB). We can therefore confirm that our use of blockchain does not result in significant amounts of "bloat."

8) Static Security Analysis

The SAST module utilizes either Slither or Mythril to automatically detect potential vulnerabilities in the codebase. It then uses a formula based on the number of high-, medium-, low-, and informational-severity issues found by Slither/Mythril to compute a composite security score:

$$\text{score} = 100-(10\times n_{\text{high}}+5\times n_{\text{med}}+2\times n_{\text{low}}+1\times n_{\text{info}}) \quad (14)$$

If one or both of the SAST tools is missing, fails, or is unavailable, the module will simply skip that step, thus making it possible to execute tests in a variety of different test environments.

9) Summary of Blockchain Validations

A summary of the blockchain validations is provided in Table 23.

VIII. DISCUSSION

A. COMPARABILITY GLOBALLY WITH ENHANCED OPERATIONALLY INTERPRETABLE OUTPUT

Linear Regression continues to be nearly tied globally (i.e., comparable) to other methods as the global baseline due to the high degree of determinism present within the dataset. There are both physical and data-centric reasons for this. Physically, the solar production pattern in Tamil Nadu at the hourly aggregate scale is dictated by recurring astronomical and seasonal patterns, whereas aggregated statewide wind (which can be less predictable at site-specific levels) is generally smoother. Data-centrically, the application of statewide aggregation, as well as the engineering of cyclical features, makes the regularity of the data explicit.

An essential mechanistic reason that explains the relatively direct algebraic relationship between the input features and the target variable is that P_t^{gen} is directly part of the feature vector and y_t is a monotonically increasing function

TABLE 24. Root-Cause Analysis of Standalone BiLSTM Underperformance

Issue	Why It Hurts BiLSTM	Confidence
Target near-linear	Algebraic closeness between P^{gen} and target saturates nonlinear signal; capacity wasted on deterministic structure	High
No long memory needed	Diurnal structure is periodic ($\sim$24 h); recurrent state beyond 2 h provides diminishing returns	Medium-High
Attention overfits	With modest data, attention weights overfit training distribution rather than generalizing to unseen peaks	Medium
Modest data size	6,084 training sequences constrain parameterized nonlinear model vs. analytical linear fit	High

of P_t^{gen}/P^{peak}. Thus, Linear Regression establishes a near-linear relationship. As such, this relationship also represents all of the potential signal for non-linear learning models to learn about the target variable using all of the available input features.

As per Occam's Razor principle, if two forecasting models are approximately equally effective in terms of forecasting accuracy, then the simpler model should be selected as the baseline. However, unlike simple Linear Regression, the Hybrid Residual model retains a large portion of the original deterministic structure while only selectively modeling the residual component that accounts for non-linearity. Therefore, it is selective complexity—not additional complexity—that defines the contribution.

B. STRUCTURAL ANALYSIS OF WHY STAND-ALONE BI-LSTMS FAILED TO OUT-PERFORM BASELINE MODELS

Table 24 lists the top four hypothesized explanations for why stand-alone BiLSTMs failed to perform better than baseline models on this task, in addition to a brief discussion of each rationale and the confidence in each hypothesis. These findings provide evidence to support practitioners' claims that poor performance was due to a structural mismatch between task requirements and model capabilities (rather than solely due to insufficient training data or poorly chosen hyperparameters).

C. ERROR VARIANCE REDUCTION

The residual analysis indicates that the linear baseline produced heteroscedastic errors (greater variance at higher generation levels). Peak renewable generation hours represent those most likely to prompt scheduling, trading, or storage decisions. The proposed framework addresses this through the combination of a reliable deterministic base model with a secondary residual component providing focused correction capacity.

D. WHY NOT TRANSFORMERS?

Transformers are powerful but typically require more data, memory, and computation than lightweight recurrent alternatives [36], [38]–[40]. Standard Transformer self-attention exhibits quadratic memory growth with sequence length. For this scale (8760 observations, 6084 training sequences), a 4,081-parameter attention-BiLSTM is more efficient and easier to deploy near the edge. Future work could evaluate Transformer or state-space sequence models as alternative residual learners while preserving the same decomposition principle.

E. INTERPRETABILITY

Residual decomposition gives a type of architectural interpretability that standalone deep models cannot offer. Linear regression captures the deterministic scaffolding (diurnal cycle, capacity factor trend, load correlation) while the attention-BiLSTM captures nonlinear residuals (weather-driven variability, ramp events, peak-hour anomalies). Separating the two allows practitioners to debug by examining the quality of fit for both portions independently. A practitioner can trace back the individual components of a PGS signal—the dominant linear portion and the bounded BiLSTM adjustment can be audited individually.

F. LIMITATIONS

This study uses only a single regional case and one year of hourly data. The modeled loads and batteries provide an excellent testing ground but further research should examine real-time operational loads and storage telemetry. The blockchain evaluation was performed on a local Hardhat node (instant mining for automine, configurable interval mining); deployment on a live network would require analysis of variable gas pricing, network congestion, mempool dynamics, and privacy. Future research should quantify uncertainty for forecast-gated blockchain execution, including prediction interval estimation and confidence score calibration.

G. BROADER METHODOLOGICAL LESSON

Strong deep-learning architectures do not automatically outperform simpler models on structured grid data. Three progressively elaborated standalone BiLSTM architectures—each incorporating additional attention complexity—all failed to match a simple linear baseline. The crucial insight was not to keep making the BiLSTM larger but to ask what the model should actually be learning. Once the question was reframed from "how do we improve standalone BiLSTM performance?" to "what signal remains after the linear model has explained what it can?", the path to a high-performing architecture became clear. This reframing is broadly applicable.

H. MULTI-SITE GENERALIZABILITY

To assess whether the findings generalize beyond Tamil Nadu, the same pipeline is applied to 9 additional Indian states (GJ, RJ, KA, MH, AP, MP, OR, UP, WB) that have both solar and wind capacity-factor data. Table 25 reports the LR and Hybrid Residual RMSE with Diebold-Mariano significance for each site.

Key observations from the multi-site evaluation:

- Across all 10 states, LR achieves mean RMSE of 0.007787 ± 0.001608 and mean R^2 of 0.994791, confirming that the linear baseline is consistently strong for state-level hourly renewable aggregation.
- Across 10 Indian states, the proposed Hybrid Residual model achieved performance comparable to Linear Regression (mean RMSE: 0.008626 vs. 0.007787). Diebold–Mariano analysis supported similar predictive accuracy in most states (mean $p = 0.19$), while Gujarat (GJ) was the only state where the Hybrid Residual model consistently demonstrated a statistically significant advantage across all five random seeds ($p < 0.05$).
- WB shows instability in one seed (std $= 0.0107$), suggesting that the Hybrid Residual model can occasionally diverge in states with different generation profiles, reinforcing the safety-net property of the residual architecture (worst-case reverts toward LR).
- The near-parity finding is not specific to Tamil Nadu but reflects the structural characteristics of hourly state-level renewable aggregation across diverse Indian grid regions.

IX. CONCLUSION

This paper presents a residual decomposition approach for renewable-energy forecasting and blockchain-anchored scheduling. The model decomposes the generation series into deterministic and stochastic components, using Linear Regression as the baseline and an attention-based BiLSTM to learn residual errors. The two-stage design was chosen based on empirical evidence: all standalone attention-based models underperformed compared with Linear Regression, and residual error analysis indicated that nonlinear errors were concentrated in a small fraction of test points.

Evaluation across five random seeds with Diebold-Mariano significance testing shows that the Hybrid Residual model achieves near-parity with Linear Regression on global metrics (RMSE 0.006918 ± 0.001294 vs. LR RMSE 0.006333; $R^2 = 0.985837 \pm 0.005798$ vs. LR $R^2 = 0.988452$) without statistically significant difference ($p > 0.05$ on both full-test and peak-test subsets). The peak subset ($y \geq 0.20$) contains 114 of 1266 test points (9.0%).

Relative to Random Forest, the Hybrid Residual model reduces RMSE by 18.6% and unexplained variance by 33.7%, and outperforms all standalone deep-learning architectures tested (LSTM, GRU, BiLSTM, XGBoost). Most importantly, the residual decomposition preserves the performance of the Linear Regression baseline while providing an interpretable, decomposed architecture suitable for downstream control applications.

The scheduling results show that a 0.15 threshold selects 280 of 1266 hours and increases renewable share from 15.34% to 25.81% during scheduled windows (a 68.26% relative improvement). However, because LR and the Hybrid Residual model produce identical scheduling outcomes under this rule, the operational gain is attributed to thresholded selection rather than to a demonstrated Hybrid Residual forecast advantage.

The blockchain component is validated through a 12-module testing suite comprising 62 automated test cases (all passing) that evaluate hash integrity, chain linkage, PGS gating correctness, gas economics, temporal performance, throughput, storage efficiency, and static security analysis. The prototype achieves 709 TPS for forecast anchoring, 2.47 ms end-to-end pipeline latency on automine, and a 2.24:1 storage compression ratio through the hybrid on-chain hash/off-chain payload design. All PGS boundary enforcement and tamper detection tests pass with 100% accuracy. These results confirm that the oracle architecture is functionally correct and computationally feasible. Production deployment on mainnet or L2 rollups, privacy-preserving mechanisms, and adversarial robustness evaluation remain future work.

X. FUTURE WORK

Several directions can extend the present framework:

1) Transformers and state-space sequence models (S4, Mamba) could be used as alternative residual learners while maintaining the same decomposition principle.
2) Adaptive attention with weather-regime conditioning may improve performance during meteorological edge cases.
3) Real-time deployment using rolling retraining will validate whether this pipeline remains accurate as the Tamil Nadu grid mix evolves.
4) Additional physical modeling combining demand telemetry and energy price signals will allow full end-to-end validation in a production environment.
5) Uncertainty quantification for forecast-gated blockchain execution—including prediction intervals and calibrated confidence scores into the PGS computation—will allow the oracle to distinguish high-confidence from low-confidence scheduling windows.
6) Deployment of the smart contract prototype on Ethereum L2 rollups (Optimism, Arbitrum) or application-specific chains with empirical measurement of mainnet gas costs, network latency under congestion, and privacy-preserving mechanisms (e.g., zero-knowledge proofs for payload confidentiality).

DECLARATIONS

Funding: The authors declare that no external funding was received for this work.

Conflicts of Interest: The authors declare no conflicts of interest.

Data Availability: The raw renewable resource data are available from the GridPath India dataset on Dryad [64]. Processed project artifacts are maintained in the project workspace under `raw_data/raw_data`.

Ethical Approval: This study uses publicly available energy-system data and does not involve human participants or animal subjects.

TABLE 25. Multi-Site Evaluation: LR vs Hybrid Residual Across 10 Indian States (Log-Scale, 5-Seed)

State	n_{test}	n_{peak}	LR RMSE	Hybrid Residual RMSE (mean±std)	LR R^2	Hybrid Residual R^2	DM $\bar{p}$	Sig. seeds
TN	1308	303	0.010753	0.011020 ± 0.000607	0.989088	0.988511	0.2859	1/5
GJ	1308	365	0.006175	0.006163 ± 0.000034	0.997224	0.997235	0.0010	5/5
RJ	1308	333	0.006325	0.006335 ± 0.000008	0.996806	0.996796	0.1773	1/5
KA	1308	356	0.009016	0.009033 ± 0.000027	0.993713	0.993688	0.2956	2/5
MH	1308	352	0.006282	0.006294 ± 0.000005	0.997106	0.997095	0.0627	3/5
AP	1308	377	0.007139	0.007152 ± 0.000023	0.996258	0.996245	0.2368	1/5
MP	1308	336	0.007368	0.007372 ± 0.000016	0.995773	0.995769	0.1278	4/5
OR	1308	345	0.007080	0.007284 ± 0.000469	0.996534	0.996319	0.2313	2/5
UP	1308	251	0.009996	0.009995 ± 0.000019	0.990262	0.990263	0.4499	1/5
WB	1308	312	0.007732	0.015611 ± 0.010737	0.995145	0.972724	0.0807	2/5
Mean	1308	333	0.007787	0.008626 ± 0.002960	0.994791	0.992464	0.1949	–

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